DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON

SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
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ARGIHOME

SAGAR KHADKA
ALEX TABLIZO
GOVINDA AWASTHI
LAXMAN BHUSHAL
MUHAMMAD SAAD MEMON
NATHAN NGUYEN TRAN
THUC DANG

Revision History

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0.1	10.01.2015	GH	document creation
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0.3	10.12.2015	AT, GH	release candidate 1
1.0	10.20.2015	AT, GH, CB	official release
1.1	10.31.2015	AL	added customer change requests

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1 Product Concept

This section describes the purpose, intended use, and target users of the Agrihome product. Agrihome is a smart indoor plant monitoring and early disease detection system designed to help users observe plant health before visible damage becomes severe. The system combines image capture, environmental sensing, backend analysis, and a user-facing dashboard to provide health insights, warnings, and historical tracking for home and small-scale growers.

1.1 Purpose and Use

Agrihome is intended to support plant owners by continuously monitoring plant condition and identifying potential health problems as early as possible. The product focuses on indoor use and is especially suited for crops or household plants that benefit from regular observation, such as tomatoes and other small garden plants.

The system is designed to capture plant images from one or more cameras and collect supporting environmental data such as temperature, humidity, soil moisture, and light level when sensors are available. Captured data is analyzed by the software system to classify the plant as healthy or potentially unhealthy, and to identify likely issues such as leaf disease symptoms or environmental stress. The result is then presented to the user through a dashboard or application interface.

In normal use, a user places the plant within the monitoring area, powers the system, and connects to the dashboard. Agrihome then performs scheduled monitoring without requiring the user to manually inspect the plant each time. When the system detects abnormal plant conditions, it provides a warning and a short explanation so the user can take action earlier. The product is intended to reduce guesswork, improve response time to plant problems, and make plant care easier for non-expert users.

Agrihome is not intended to physically treat the plant, apply chemicals, or replace professional agricultural diagnosis. Instead, it acts as an intelligent monitoring and decision-support tool for plant care in home or small-scale environments.

1.2 Intended Audience

The primary intended audience for Agrihome includes home gardeners, apartment growers, student researchers, and hobbyists who want a practical way to monitor plant health with less manual effort. The product is especially useful for users who may not have advanced agricultural knowledge but still want timely feedback about plant condition.

A secondary audience includes academic teams, senior design evaluators, and small greenhouse or controlled-environment users who may use the system as a prototype platform for plant monitoring research and experimentation.

If made commercially available, Agrihome would be most attractive to consumers interested in smart gardening, indoor growing, and automated plant care support. The product is designed as a standalone monitoring system, but it could also serve as a subsystem within a larger smart agriculture or smart home platform.

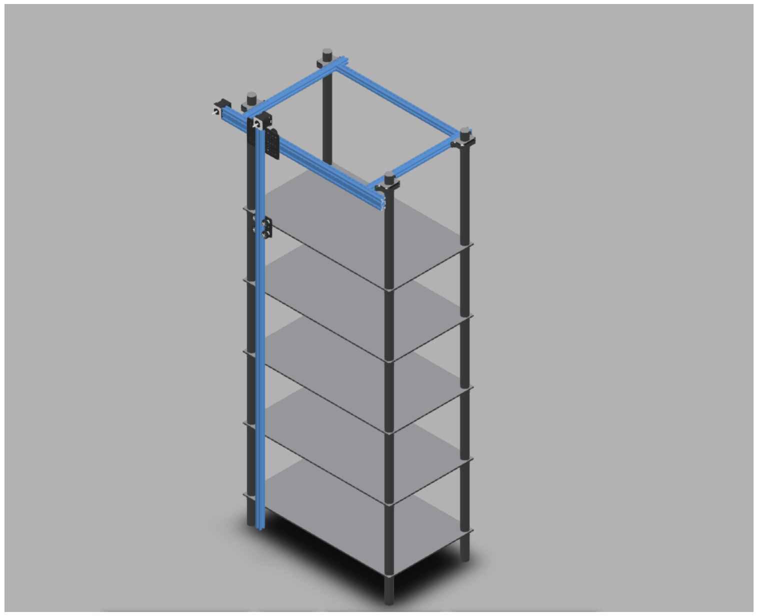

Figure 1: Agrihome conceptual overview

2 Product Description

This section provides an overview of the smart plant monitoring system, including its primary features, functions, and user interactions. The system is designed to monitor plant health using an integrated camera and environmental sensors, and to provide actionable insights through a user interface. From the perspective of end users, the system enables real-time monitoring and feedback. For maintainers and developers, it provides a structured backend for data processing and analysis. The key components, system behavior, and user interactions are described below.

2.1 Features & Functions

The smart plant monitoring system is designed to capture plant images and environmental data, process this information, and provide users with plant health insights.

The system consists of the following core components:

- **Camera Module:** Captures images of the plant at regular intervals.
- **Processing Unit:** Processes captured data and determines plant health status using backend logic.
- **Backend Server:** Stores data and performs analysis.
- **User Interface (Dashboard):** Displays plant health status, historical data, and alerts.

The system performs the following functions:

- Captures plant images automatically
- Processes data to determine plant health status
- Stores data for historical tracking
- Displays results and alerts to the user

The system does **not** perform the following:

- It does not physically interact with or modify the plant (no watering or actuation)
- It does not provide medical or agricultural guarantees
- It does not operate outdoors in extreme environmental conditions

External elements include the Internet (for data transfer) and optional cloud storage services.

2.2 External Inputs & Outputs

The system receives input from both users and hardware components, and produces outputs for user interpretation.

Name	**Description**	**Use**
Camera Image	Image of plant captured periodically	Used for plant health analysis
User Input	User settings (e.g., thresholds)	Configures system behavior
Health Status	Computed plant condition (e.g., healthy/unhealthy)	Displayed to user
Alerts	Notifications for abnormal conditions	Warns user of issues
Historical Data	Stored past readings and images	Used for tracking trends

2.3 Product Interfaces

The system provides a user-facing dashboard interface and internal system interfaces.

User Interface:

- Displays plant health status (e.g., healthy, warning, critical)
- Provides historical graphs of plant conditions
- Displays alerts and notifications

Hardware Interface:

- Camera connected to processing unit
- Sensors connected via microcontroller (e.g., Arduino or Raspberry Pi)

Software Interface:

- Backend server processes incoming data
- Communication via APIs between hardware and backend
- Optional cloud database for storage

3 Customer Requirements

This section defines the customer requirements for Agrihome. Customer requirements are the visible features and functions that the intended users, sponsor, and evaluators expect the system to provide. These requirements describe what end users should expect the product to do during normal operation, what information the product should present, and how the user should interact with the system. Because Agrihome is intended to monitor plant health using image capture, environmental sensing, backend analysis, and a dashboard interface, the requirements in this section focus on observable monitoring behavior, user interaction, alerts, and historical tracking.

3.1 Plant Health Status Display

3.1.1 Description

The system shall provide a user-facing dashboard that displays the current health status of the monitored plant. At minimum, the dashboard shall show a plant health result such as healthy, warning, or critical, along with the latest available monitoring result for the user to review. This requirement ensures that users can directly observe the main output of the product.

3.1.2 Source

Project Charter, sponsor expectations, and product description.

3.1.3 Constraints

The dashboard must remain simple enough for non-technical users while still showing meaningful monitoring results. The design is constrained by available frontend development time and prototype scope.

3.1.4 Standards

The interface should follow common dashboard usability and readability practices for software systems.

3.1.5 Priority

Critical

3.2 Automatic Image Capture

3.2.1 Description

The system shall automatically capture plant images at regular intervals for monitoring purposes. This requirement supports the main product goal of reducing the need for constant manual inspection by the user.

3.2.2 Source

Product description and project concept.

3.2.3 Constraints

Image capture reliability depends on camera hardware, lighting conditions, plant position, and available processing and storage resources.

3.2.4 Standards

The image capture process should follow normal hardware operating specifications provided by the selected camera manufacturer.

3.2.5 Priority

Critical

3.3 Environmental Data Collection

3.3.1 Description

The system shall collect environmental sensor data associated with plant monitoring when sensors are available. This data may include measurements such as temperature, humidity, soil moisture, or related environmental conditions used by the system for monitoring and analysis.

3.3.2 Source

Product description and hardware interface requirements.

3.3.3 Constraints

This requirement depends on successful sensor integration, calibration, and communication with the processing unit. Sensor quality and hardware cost may affect the completeness and accuracy of collected data.

3.3.4 Standards

Sensor components should follow manufacturer specifications and standard embedded sensor interfacing practices.

3.3.5 Priority

High

3.4 Alert Display

3.4.1 Description

The system shall display alerts or notifications when abnormal plant conditions are detected. Alerts shall be visible to the user through the dashboard and shall indicate that the plant may require attention.

3.4.2 Source

Product description, customer need, and sponsor expectations.

3.4.3 Constraints

Alert usefulness depends on the quality of captured images, sensor readings, and analysis results. Excessive false alerts may reduce user trust in the system.

3.4.4 Standards

The alert system should follow general software notification practices for clear wording and visible warning presentation.

3.4.5 Priority

Critical

3.5 Historical Data Display

3.5.1 Description

The system shall allow the user to review historical monitoring data, including past readings and captured images when available. This feature shall support tracking changes in plant condition over time.

3.5.2 Source

Product description and intended user needs.

3.5.3 Constraints

History depth may be limited by storage capacity, image size, monitoring frequency, and database design choices within the prototype.

3.5.4 Standards

Stored monitoring data should follow standard timestamping and record consistency practices for software systems.

3.5.5 Priority

High

3.6 User Settings Input

3.6.1 Description

The system shall accept user input for configuration settings such as monitoring thresholds or related behavior settings. This requirement allows the user to adjust the system to better match specific monitoring preferences or plant conditions.

3.6.2 Source

Product description and expected user interaction.

3.6.3 Constraints

The number of configurable options may be limited by implementation time, user interface simplicity, and backend support for settings management.

3.6.4 Standards

User configuration features should follow common software input validation and settings management practices.

3.6.5 Priority

High

3.7 Dashboard Interface

3.7.1 Description

The system shall provide a dashboard-based user interface through which users can view plant health status, alerts, and historical monitoring information. This dashboard shall serve as the primary interaction point between the user and the system.

3.7.2 Source

Product description and project design goals.

3.7.3 Constraints

Interface complexity must be limited enough to remain understandable to non-expert users. Time and frontend development resources may limit visual polish in the prototype.

3.7.4 Standards

The interface should follow standard usability and readability practices for browser-based dashboard systems.

3.7.5 Priority

Critical

3.8 No Physical Plant Actuation

3.8.1 Description

The system shall not physically interact with or modify the plant. The product shall not perform watering, spraying, fertilizing, or any other automatic physical actuation on the plant. This requirement clearly defines the scope of the prototype as a monitoring and analysis system only.

3.8.2 Source

Product description and project scope definition.

3.8.3 Constraints

This requirement limits the system to sensing, processing, storage, and display functions only. Any future physical automation would require additional hardware, controls, and safety considerations.

3.8.4 Standards

The requirement follows the documented functional scope of the prototype system.

3.8.5 Priority

Critical

3.9 Remote Notification Support

3.9.1 Description

In a future version, the system should support remote notification delivery such as email, mobile push notification, or text message when abnormal plant conditions are detected. This would allow users to receive alerts even when not actively viewing the dashboard.

3.9.2 Source

Future customer convenience needs and product expansion goals.

3.9.3 Constraints

This feature would require additional backend services, notification platform integration, secure internet communication, and user account support. It may not be feasible within the current prototype scope.

3.9.4 Standards

Future implementations should follow secure communication practices and standard authentication methods.

3.9.5 Priority

Future

4 Packaging Requirements

The finished prodect will include an overhead light fixture with the camera rig installed, a web client accessable on any device with a web browser, and access to AI image processing via a cloud server. Packaging for the camera rig will require significant assembly time. Since the light fixture is designed to mount to any pre-exisiting shelf, the user must be able to adjust the mounting points of the rig to attach it to their shelf.

4.1 Hardware Packaging

4.1.1 Description

Hardware for the camera system will be partally assembled, but mounting the light fixture will require adjustment by the user. The hardware package will include custom mounting hardware for their shelf.

4.1.2 Source

Project Charter

4.1.3 Constraints

Size of packaging, and complexity of assembly will restrict the amount of neccessary pre assembled parts.

4.1.4 Standards

None

4.1.5 Priority

Medium

4.2 Software Packaging

4.2.1 Description

Access to the main control interface for the plant detection system will be through a web server accessable through a web browser. All AI image processing is done on the cloud, and all image capture is done on an included computer. No additional software should be required.

4.2.2 Source

Project Charter

4.2.3 Constraints

Processing power of the onboard computer limits the amount of image processing done on it, as well as its ability to host a web server.

4.2.4 Standards

None

4.2.5 Priority

Medium

5 Performance Requirements

Preformance requirements for the plant monitor include are reliability of the system, and the speed of scanning, and image processing

5.1 Reliability

5.1.1 Description

Reliabilty refers to the both image capture and AI detection. The system for the image capture must be able to take consistant pictures of leaves with acceptable quality, lighting, and angle. The AI algorithm must detect any signs of disease with high accuracy without triggering false alarms. Every step of image capture and processing must be done without human intervention.

5.1.2 Source

Project Charter

5.1.3 Constraints

Constraints include image quality, lighting quality, and angle of leaves relative to the camera.

5.1.4 Standards

None

5.1.5 Priority

High

5.2 Speed

5.2.1 Description

Speed refers to the time it take from initializing image capture, to a result sent to the web client. A low time between initialize and result provides a smoother experience for the user and less time when they are unable to physically interact with the plant rack.

5.2.2 Source

Project Charter

5.2.3 Constraints

Constraints include processing speed, network speeds, and time for image capturing by the camera rig.

5.2.4 Standards

None

5.2.5 Priority

Low

6 Safety Requirements

This section defines the safety requirements for the smart plant monitoring system. The system includes a camera mounted on a stand, sensors, and backend processing components. Safety considerations focus on preventing electrical hazards, ensuring safe physical design, and protecting users from environmental or operational risks. The product must be safe for indoor use, especially in home or laboratory environments, and must not expose users to harmful materials, electrical shocks, or unsafe mechanical structures.

6.1 Laboratory equipment lockout/tagout (LOTO) procedures

6.1.1 Description

Any fabrication equipment provided used in the development of the project shall be used in accordance with OSHA standard LOTO procedures. Locks and tags are installed on all equipment items that present use hazards, and ONLY the course instructor or designated teaching assistants may remove a lock. All locks will be immediately replaced once the equipment is no longer in use.

6.1.2 Source

CSE Senior Design laboratory policy

6.1.3 Constraints

Equipment usage, due to lock removal policies, will be limited to availability of the course instructor and designed teaching assistants.

6.1.4 Standards

Occupational Safety and Health Standards 1910.147 - The control of hazardous energy (lockout/tagout).

6.1.5 Priority

Critical

6.2 National Electric Code (NEC) wiring compliance

6.2.1 Description

Any electrical wiring must be completed in compliance with all requirements specified in the National Electric Code. This includes wire runs, insulation, grounding, enclosures, over-current protection, and all other specifications.

6.2.2 Source

CSE Senior Design laboratory policy

6.2.3 Constraints

High voltage power sources, as defined in NFPA 70, will be avoided as much as possible in order to minimize potential hazards.

6.2.4 Standards

NFPA 70

6.2.5 Priority

Critical

6.3 RIA robotic manipulator safety standards

6.3.1 Description

Robotic manipulators, if used, will either housed in a compliant lockout cell with all required safety interlocks, or certified as a "collaborative" unit from the manufacturer.

6.3.2 Source

CSE Senior Design laboratory policy

6.3.3 Constraints

Collaborative robotic manipulators will be preferred over non-collaborative units in order to minimize potential hazards. Sourcing and use of any required safety interlock mechanisms will be the responsibility of the engineering team.

6.3.4 Standards

ANSI/RIA R15.06-2012 American National Standard for Industrial Robots and Robot Systems, RIA TR15.606-2016 Collaborative Robots

6.3.5 Priority

Critical

7 Security Requirements

Include a header paragraph specific to your product here. In this section specify any requirements related to information security or privacy. This may include items such as encryption standards, data storage procedures, authentication, password strength, etc.

7.1 Requirement Name

7.1.1 Description

Detailed requirement description...

7.1.2 Source

Source

7.1.3 Constraints

Detailed description of applicable constraints...

7.1.4 Standards

List of applicable standards

7.1.5 Priority

Priority

8 Maintenance & Support Requirements

Include a header paragraph specific to your product here. Maintenance and support requirements address items specific to the ongoing maintenance and support of your product after delivery. Think of these requirements as if you were the ones who would be responsible for caring for customers/end user after the product is delivered in its final form and in use "in the field". What would you require to do this job? Specify items such as: where, how and who must be able to maintain the product to correct errors, hardware failures, etc.; required support/troubleshooting manuals/guides; availability/documentation of source code; related technical documentation that must be available for maintainers; specific/unique tools required for maintenance; specific software/environment required for maintenance; etc.

9 Maintenance Requirements

This section defines the maintenance and support requirements for the Agrihome smart plant monitoring system after deployment. Maintenance requirements ensure that the system can be updated, repaired, and supported over time to maintain reliability and usability. These requirements address hardware servicing, software updates, troubleshooting procedures, documentation availability, and system support responsibilities. The goal is to ensure that both developers and users can effectively maintain system operation and resolve issues when they arise.

9.1 System Maintenance Documentation

9.1.1 Description

The system shall provide complete maintenance documentation including setup guides, troubleshooting instructions, and system architecture details. This documentation shall allow developers and maintainers to understand system components, diagnose issues, and perform updates or repairs efficiently.

9.1.2 Source

Project design requirements and maintainability best practices.

9.1.3 Constraints

Documentation must be kept updated as system changes occur. Limited development time may restrict the level of detail included in the initial version.

9.1.4 Standards

Documentation should follow standard software engineering documentation practices.

9.1.5 Priority

High

9.2 Hardware Maintenance and Replacement

9.2.1 Description

The system shall allow easy access to hardware components such as the camera module, sensors, and processing unit for repair or replacement. Components should be replaceable without requiring complete system disassembly.

9.2.2 Source

Hardware design considerations and user maintenance needs.

9.2.3 Constraints

Physical design limitations and mounting structure may restrict accessibility. Replacement parts must be compatible with the system.

9.2.4 Standards

Hardware components should follow manufacturer specifications and safe handling practices.

9.2.5 Priority

Critical

9.3 Software Updates and Bug Fixes

9.3.1 Description

The system shall support software updates to fix bugs, improve performance, and enhance features. Updates should be deployable without requiring complete system reinstallation.

9.3.2 Source

Software lifecycle management requirements.

9.3.3 Constraints

Update process depends on network availability and system configuration. Improper updates may affect system stability.

9.3.4 Standards

Software updates should follow standard version control and deployment practices.

9.3.5 Priority

Critical

9.4 System Monitoring and Error Logging

9.4.1 Description

The system shall maintain logs of system activity, errors, and failures to assist in troubleshooting and maintenance. Logs should include timestamps and relevant system events.

9.4.2 Source

System reliability and debugging requirements.

9.4.3 Constraints

Storage limitations may restrict the amount of log data retained. Logging must not significantly affect system performance.

9.4.4 Standards

Logging should follow common software debugging and monitoring practices.

9.4.5 Priority

High

9.5 Technical Support Availability

9.5.1 Description

The system shall provide access to technical support resources such as user guides, FAQs, or developer support documentation to assist users in resolving issues.

9.5.2 Source

Customer support and usability requirements.

9.5.3 Constraints

Support availability may be limited based on development team resources. Real-time support may not be feasible in the prototype stage.

9.5.4 Standards

Support materials should follow standard customer support documentation practices.

9.5.5 Priority

Medium

10 Other Requirements

Include a header paragraph specific to your product here. In this section specify anything else that is required for the product to be deemed complete. Include requirements related to customer setup and configuration if not specified in a previous requirement. Add any known requirements related to product architecture/design, such as modularity, extensibility (for future enhancements), or adaptation for a specific programming language. Consider requirements such as portability of your source code to various platforms (Windows, Linux, Unix Mac OS, etc.).

10.1 Requirement Name

10.1.1 Description

Detailed requirement description...

10.1.2 Source

Source

10.1.3 Constraints

Detailed description of applicable constraints...

10.1.4 Standards

List of applicable standards

10.1.5 Priority

Priority

11 Future Items

This section summarizes features and functionalities that were considered during the design of the smart plant monitoring system but will not be implemented in the current prototype due to limitations in time, resources, and technical complexity. These features may be included in future versions of the system.

11.1 Mobile application support

11.1.1 Description

A dedicated mobile application will allow users to monitor plant health, receive notifications, and control system settings remotely.

11.1.2 Source

User experience enhancement goals

11.1.3 Constraints

Requires additional development time, cross-platform compatibility, and backend integration.

11.1.4 Standards

Mobile application development standards (iOS/Android guidelines)

11.1.5 Priority

Future

11.2 Advanced AI-based plant disease detection

11.2.1 Description

The system will use machine learning models to detect specific plant diseases from captured images and provide detailed diagnostics and recommendations.

11.2.2 Source

Project design discussion

11.2.3 Constraints

Requires large datasets, model training, and higher computational resources.

11.2.4 Standards

AI/ML best practices and data handling guidelines

11.2.5 Priority

Future

11.3 Cloud-based data analytics and sharing

11.3.1 Description

The system will support cloud integration for storing plant data, enabling long-term analytics and sharing data across multiple devices.

11.3.2 Source

System scalability considerations

11.3.3 Constraints

Requires cloud infrastructure, security implementation, and ongoing maintenance.

11.3.4 Standards

Cloud computing and data security standards

11.3.5 Priority

Future

References